

Day 38: Level Order Traversal

- ◆ Level Order Traversal kya hota hai?

Is traversal me hum tree ko level by level traverse karte hain:

- 👉 Pehle root node
- 👉 Phir uske children
- 👉 Phir unke children (level-wise)

Example:

```
    1
   /\
  2 3
 /\  \
4 5 6
```

- 👉 Output:

1 2 3 4 5 6

-
- ◆ Kaunsa Data Structure use hota hai?

- 👉 Queue (FIFO) use hota hai

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- ◆ Important Points (Interview Ready)

- ✓ BFS based traversal hai
- ✓ Queue use hota hai
- ✓ Time Complexity $\rightarrow O(n)$
- ✓ Space Complexity $\rightarrow O(n)$ (Queue ke liye)

◆ Java Code (Complete Program with Output)

```

1. package Day_38;
2.
3. import java.util.*;
4.
5. class LevelOrderTraversal {
6.
7.     static class Node {
8.         int data;
9.         Node left, right;
10.
11.         Node(int data) {
12.             this.data = data;
13.             left = right = null;
14.         }
15.     }
16.
17.     Node root;
18.
19.     // Level Order Traversal Function
20.     public void levelOrder() {
21.         if (root == null) return;
22.
23.         Queue<Node> q = new LinkedList<>();
24.         q.add(root);
25.
26.         while (!q.isEmpty()) {
27.             Node temp = q.poll();
28.             System.out.print(temp.data + " ");
29.
30.             if (temp.left != null) {
31.                 q.add(temp.left);
32.             }
33.
34.             if (temp.right != null) {
35.                 q.add(temp.right);
36.             }
37.         }
38.     }
39.
40.     public static void main(String[] args) {
41.         LevelOrderTraversal tree = new LevelOrderTraversal();
42.
43.         // Tree creation
44.         tree.root = new Node(1);
45.         tree.root.left = new Node(2);
46.         tree.root.right = new Node(3);
47.         tree.root.left.left = new Node(4);
48.         tree.root.left.right = new Node(5);
49.         tree.root.right.right = new Node(6);
50.
51.         System.out.println("Level Order Traversal:");
52.         tree.levelOrder();
53.     }
54. }
55.

```

◆ Output:

Level Order Traversal:

1 2 3 4 5 6